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Applying Information Technology To Organization Design

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In the past, organization theory has been mainly concerned with what might be called “organization for production”—that is, with systems that use the services of substantial numbers of employees to generate, more or less continuously, some kind of output or “product.”¹ The normative theory of organization, aimed at enhancing organizational efficiency and effectiveness, traditionally paid special attention to two problems: how to divide up the work for its efficient performance and in such a way as to keep the needs for coordination of the parts within manageable bounds; and how to construct and maintain mechanisms for coordinating the several organizational parts—especially authority mechanisms.

Research on “human relations” in organizations, beginning on a substantial scale in the 1930s, turned attention in organizational design to the linkage between the individual as organization member and the pattern of organizational activity. The principal normative concern here was to create organizational environments in which employees would be motivated to join the organization, to remain in it, and to contribute vigorously and effectively to its goals. As a result of theory and empirical research in human relations, the factory and office came to be viewed as relatively impoverished human environments—starving both the human mind and the human emotions. Proposals were advanced for the redesign of organizations in ways that would make work intellectually more stimulating (or less boring), and give the worker a greater feeling of participation in the decisions governing his ac-

tivity. “Job enrichment” and “democratic management” are labels commonly applied to these emphases in organizational design.²

The human relations movement did not limit itself to redesigning organizations to achieve the traditional goals of organizational effectiveness and efficiency. It also raised fundamental value questions of whether organizations *should* be designed in these terms; or whether, on the other hand, there should be a deliberate sacrifice of effectiveness and efficiency in order to make work itself a rewarding and enjoyable part of human life. This value issue is sometimes obscured in the literature when the assumption is made, implicitly or explicitly, that “the happy employee is the productive employee.” That assumption tends to prevail through the early writings on human relations; but has been frequently questioned or attacked in recent years by the social critics of the New Left.

I do not intend to enter into that controversy here. I raise the issue because I think the debate has been carried out on premises that are less and less valid in today’s organizations, and will be still less valid in the organizations of the future. The attack on work in organizations as dehumanizing generally takes as its model of the organization a system engaged in mass repetitive processing of materials or symbols—the assembly line or a room full of clerks or draftsmen. Chaplin’s movie, “Modern Times,” exaggerates only slightly the portrait of the factory given in the human relations literature.

But with the introduction of highly automated machinery, and particularly with the introduction

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of mechanized information processing equipment, the assembly line becomes a rather rare form of organization of production, as does the repetitive unautomated clerical process. The human operative or clerk is less often a cog in the ongoing production process, or even the first-level controller of that process. He is more and more an observer, moderator, maintenance man, and repairman for a nearly autonomous process that can carry on for significant intervals of time without direct human intervention.

The question of whether "the happy employee is the productive employee" still needs asking under these new conditions of employment. But it is not obvious that the question will have the same answer it had in the past. The problems of cognitive impoverishment and alienation may be replaced by quite new problems—or may disappear altogether. My own assessment—based on direct observation and the studies of others of highly automated work situations—is that the new environment will be, for most workers, a more pleasant and humane environment than the old. We will need a great deal more experience with this new environment, however, to provide solid evidential foundations for that optimism.

But I am getting a bit ahead of my story, for I have not yet said why I think organizations are changing in the directions I have mentioned.

The Post-Industrial Society

Peter Drucker has used the phrase "post-industrial society" to describe the emerging world in which manufacturing production, and the activities associated with it, plays a much less central role than it did in the world of the past century. Organizations in the post-industrial society provide services, many of them intangible, more than they manufacture things. Already, a large part of the economic activity of our society consists in providing services for education, health, and leisure-time activities.

Providing services tends to pose different organizational problems from producing tangible goods. It is usually more difficult to define appropriate output measures for service organizations than for organizations that produce tangible commodities. Whatever problems are present in measuring the quality of goods are magnified greatly in measuring the quality of services. The point can be illustrated by comparing two versions

of the same economic activity, first viewed as goods-producing activity, then as a service-producing activity: that is, producing *houses* and *housing* respectively.

A house is a tangible commodity that can be manufactured and distributed through the usual market mechanisms; housing is a bundle of services provided by a dwelling in the context of a neighborhood, with schools, streets, shopping facilities, and a pattern of social interaction among the inhabitants. However complex it may be to define the qualities of a house, narrowly conceived as a structure, it is far more complex to define the qualities of housing, in the sense of a situation that creates and supports a pattern of social activity.

Related to the tendency of organizations in our society to broaden the definition of their goals from the production of tangible commodities to the production of bundles of services that may or may not be associated with tangible commodities, is a tendency to broaden their concern for the externalities associated with their activities. Externalities are simply those consequences of action that are not charged, through the existing market mechanisms, to the actors. The classical example is the factory smoke whose social costs have not generally been paid by the consumers of the factory's product.

It may be that organizations producing services usually have more and larger externalities associated with their activities than organizations producing goods; it may be that we are simply becoming more sensitive in our society to the indirect consequences of organizational activity directed toward specialized goals; it may be that, with the growth of population and technology, the actual interdependencies of organizations, and hence the externalities they cause, are becoming more extensive and significant. Whatever the reasons—and all three of those mentioned probably contribute to the trend—organizational decision making in the organizations of the post-industrial world shows every sign of becoming a great deal more complex than the decision making of the past. As a consequence of this fact, the decision-making process, rather than the processes contributing immediately and directly to the production of the organization's final output, will bulk larger and larger as *the* central activity in which the organization is engaged.

In the post-industrial society, the central problem is not how to organize to produce efficiently

(although this will always remain an important consideration), but how to organize to make decisions—that is, to process information. Until recent years, decision making was exclusively a human activity; it involved processes going on inside the human head and symbolic communication among humans. In our present world, decision making is shared between the human and mechanized components of man-machine systems, the machines being those devices we call computers. The division of labor between the human and computer components in these systems has changed steadily over the past 20 years, and we can expect it to continue to change as the sophistication of computer technology—and particularly computer programming or “software” technology—grows.

The anatomy of an organization viewed as a decision-making and information-processing system may look very different from the anatomy of the same organization viewed as a collection of people. The latter viewpoint, which is the traditional one, focuses attention on the groupings of human beings—that is, the departmentalization. The former viewpoint, on the other hand, focuses on the decision-making process itself—that is, upon the flows and transformations of symbols. If we carve an organization, conceptually, into subsystems on the basis of the principal components into which the decision-making process divides, we may, and probably will, arrive at a very different dissection than if we carve it into its departmental and sub-departmental components. Moreover, the greater the interdependencies among the departmental components, the greater will be the difference in these two ways of conceptualizing the organization.

Both of these viewpoints are useful and even essential in arriving at sound designs for organizations. In this analysis, I shall emphasize the less conventional point of view and shall discuss the decision-making process disembodied, so to speak, from the flesh-and-blood (or glass and metal, as the case may be) decision makers who actually carry out this process. Instead of watching a man or computer as information reaches him and he processes it and transmits new information in his turn, we will watch information as it flows from one man or computer to another and is transformed in the course of flow. This approach, if it has no other advantages (though I believe it does), will give us a fresh look at the design of organizations.

Two Requirements of Organizational Design

The division of labor is quite as important in organizing decision making as in organizing production, but what is being divided is different in the two cases. From the information processing point of view, division of labor means factoring the total system of decisions that need to be made into relatively independent subsystems, each one of which can be designed with only minimal concern for its interactions with the others. The division is necessary because the processors that are available to organizations, whether humans or computers, are very limited in their processing capacity in comparison with the magnitude of the decision problems that organizations face. The number of alternatives that can be considered; the intricacy of the chains of consequences that can be traced—all of these are severely restricted by the limited capacities of the available processors.

Any division of labor among decisional subsystems creates externalities, which arise out of the interdependencies among the subsystems that are ignored. What is wanted is a factorization that minimizes these externalities and consequently permits a maximum degree of decentralization of final decision to the subsystems, and a maximum use of relatively simple and cheap coordinating devices like the market mechanism to relate each of the decisional subsystems with the others.

Not only must the size of decision problems handled by organizations be reduced to manageable proportions by factorization, but the number of decisions to be processed must be limited by applying good principles of attention management. Attention management for an organization means exactly what it means for an individual human being: processing capacity must be allocated to specific decision tasks, and if the total capacity is not adequate to the totality of tasks, then priorities must be set so that the most important or critical tasks are attended to.

The information-processing systems of our contemporary world swim in an exceedingly rich soup of information, of symbols. In a world of this kind, the scarce resource is not information; it is processing capacity to attend to information. Attention is the chief bottleneck in organizational activity, and the bottleneck becomes narrower and narrower as we move to the tops of organizations, where parallel processing capacity becomes less easy to provide without damaging the coordinating

function that is a prime responsibility of these levels.

The richness of the informational environment and the scarcity of attention have many consequences for organizational design, some of which will be developed presently. At this point, only a couple of further comments need to be made. First, the difficulty of coping with the information-rich environment is compounded by the fact that most information relevant to top-level and long-run organizational decisions typically originates outside the organization, and hence in forms and quantities that are beyond its control. This means that the organization must have an "interface" for ingesting such information selectively, and for translating it into formats that are compatible with its internal information flows and systems.

Second, if attention is the scarce resource, then it becomes particularly important to distinguish between problems for decision that come with deadlines attached (real-time decisions), and problems that have relatively flexible deadlines. Rather different system designs are called for to handle these different kinds of decisions.

In summary, the inherent capacity limits of information-processing systems impose two requirements on organizational design: that the totality of decision problems be factored in such a way as to minimize the interdependence of the components; and that the entire system be so structured as to conserve the scarce resource, attention. The organizational design must provide for interfaces to handle information that originates outside the organization, and special provision must be made for decisions that have particular time limits associated with them.

Applying these basic design requirements makes it easy to see the fallacy in some recent, and more or less abortive, approaches to the improvement of information systems: municipal data banks, and management information systems. There was great enthusiasm, only a couple of years ago, for developing comprehensive data banks for metropolitan areas—these data banks to incorporate in a single system all of the myriad pieces of information about land and its uses, and about people and their activities that are generated by the operations of urban government. As the result of several attempts to construct such systems, the enthusiasm has been much moderated, and several incipient undertakings of this kind have been abandoned. There were several reasons for the

disenchantment that followed the initial attempts at constructing such systems. First, the data processing and data storage tasks proved much larger and more complex than had been imagined. Perhaps more crucial, it became less and less clear just how the data were to enter into the decision-making process, or indeed to just what decisions they were relevant.

The lesson is clear. There is no magic in "comprehensiveness." It may be sufficient motive to climb a mountain "because it is there," but the mere existence of a mass of data is not a sufficient reason for collecting it into a single, comprehensive information system. Indeed, the problem is quite the opposite: of finding way of factoring decision problems in order to relate the several components to their respective relevant data sources. Analysis of the decision-making system and its data requirements must come first; only then can a reasonable approach be made to defining the data systems that will support the decision-making process.

The history of management information systems has been nearly the same as the history of municipal data banks. In the enthusiasm to make use of the enormous power of computers, there was a tendency, in designing such systems, to take the existing source records as starting point and to try to give top management access to all this information. The question was not asked, or not asked with sufficient seriousness, whether top management either wanted or needed such information, nor whether the information that top management needed and should want could in fact be derived from these particular source records. The systems were not designed to conserve the critical scarce resource—the attention of managers—and they tended to ignore the fact that the information most important to top managers comes mainly from external sources and not from the internal records that were immediately accessible for mechanized processing.

Thus many of the efforts to design information systems for municipalities and corporations fell into the fallacy of thinking that "more information is better." They took over, implicitly, the assumptions of a past society where information rather than attention was the scarce resource.

Characteristics of the Technology

Good design requires bringing the desired ends into effective relation with the available means. To

design effective decision-making organizations, we must understand the structure of the decisions to be made; and we must understand the decision-making tools at our disposal, both human and mechanical—men and computers.

The Human Components

In our fascination with the new capabilities that computers offer us, we must be careful not to forget that our human decision makers have some pretty remarkable qualities too. Each human decision maker is provided with a sizeable memory that is stocked cumulatively over a long period of years with various kinds of relevant and irrelevant information and skills. Each is able to communicate in natural language with his fellows, either in direct face-to-face settings or by remote devices like the telephone.

Suppose, for example, that we were interested in designing an organization that would lead us to the most expert source of information in the United States about any particular question that happened to arise. Now this expert information is stored both in human heads and in books. Moreover, the information in books is also indexed in human heads, so that usually the most expeditious way to find the right book is to ask a human who is an expert on the subject the book deals with. Not only are books indexed in human heads, but people are also. Taking these resources into account, the most powerful information processing system for carrying out this search task is the aggregate of memory that is distributed among 200 million human heads, together with the telephone system that links these distributed memories. On receipt of the inquiry, I pick up the phone and call the person, among my acquaintances, whose field of expertness is as close as possible (it need not be very close at all). I ask him, not for the answer to the question, but for the name of the person in his circle of acquaintance who is closest to being an expert on the topic. I repeat the process until I have the information I want. It will be a rare instance when more than three or four calls are required.

Suppose that the question is whether whales have spleens. (I can't imagine why we want to know, but this example is as good as any other.) I call a biologist, who refers me to an ichthyologist, who refers me to a specialist on whales, who either knows the answer or can refer me to the book where I will find it.

I don't mean to propose that we junk all of

our other information systems and place sole reliance on the telephone and the vast distributed memory with which it connects us. I am even a little reluctant to mention this particular technique publicly, because if it attains general popularity, we will all be busy answering the phone, and the trick will be spoiled for those of us who use it extensively. It does provide, however, a useful illustration of how we must think about information-processing systems—including human systems—their components and interconnections, if we are to design them well. We must learn to characterize them in terms of the sizes of their memories, the ways in which those memories are indexed, their processing rates, and the rapidity with which they can respond. The human components of information systems are just as describable as the machine components, and since World War II we have learned a great deal, through psychological research, about the parameters of the human system.

Our new and growing understanding of information processing enables us to look at familiar processing systems—man and telephone—in new ways. It also introduces us to new kinds of systems, which we put under the general label of “computers,” that have capabilities of the most varied kinds.

The Computer as Memory

The computer is, first of all, a memory. I have already expressed my qualms about confusing the design of an information-*collecting* system with the design of an information-*processing* system. The fault, of course, is not in collecting information (although that may be costly in itself); it is in demanding the scarce attention of decision makers to the information that has been collected. Memories, as components of information-processing systems, need to be viewed as stores of *potential* information, which, if indexed effectively, can become available at a reasonable cost whenever it is needed as input to a decision-making process.

Consider a man who has collected a library of 30,000 books. Even if he reads one book a day—a pretty good clip—it will take him 100 years to read through all the shelves. We may even consider it a bit ostentatious of him to have collected more books than he can possibly read—as though he were trying to impress us with his learning. However, we must not be too hasty in judging him. If his library is properly indexed, then our collector has potential access to *any* of the

information in the 30,000 volumes. He is quite justified in collecting more volumes than he can read if he cannot predict in advance what particular information he will need in the future.

The computer memories that are employed today are not, in general, large compared with the paper-and-ink memories we call libraries. They are, in general, better indexed for rapid retrieval of information, and one of the important directions of technological progress since the computer has appeared on the scene has been our understanding of the indexing and information retrieval processes, and in our ability to carry these out mechanically.

The Computer as Processor

In addition to being a memory, the computer is also a processor that possesses quite general capabilities for handling symbols of all kinds, numerical and non-numerical. This is the computer's most novel feature. Nonhuman memories have been familiar to man since the invention of writing. Nonhuman symbol manipulation is something quite new, and even after 20 years, we are just beginning to glimpse its potential.³

Up to the present time, perhaps the most important use of the computer in decision making (though not the use that accounts for the bulk of computer time that is consumed by organizations) is to model complex situations, and to infer the consequences of alternative decisions. Some of this modelling makes use of mathematical techniques, like linear programming, that permit the calculation of optimal courses of action, hence serve as direct decision-making tools. In other forms of modelling, the computer serves as a simulator, calculating out the alternative histories of a system that would follow on different decision strategies.

The term "management information system" has generally been construed narrowly, and has been applied to large information storage and retrieval systems, like those mentioned earlier, in which the computer does only very simple processing of the information. The term would be better applied to the optimizing and simulation models that are increasingly used to illuminate various areas of management decision—models that are usually referred to as "operations research" and "strategic planning." Such models, however they are labelled, probably give us a better preview of the future uses of computers in organizational decision systems than do the explicitly named

management information systems.

Let me cite one example of an area of application for a strategic planning model. In the next decades, our society faces some important and difficult policy decisions with respect to the production and use of energy. In the past, the national energy problem was perceived mostly as a resource problem, and it was left in considerable part to private management through market mechanisms. Today, we see that the use of energy has important indirect consequences for the environment, and we see also that the adequacy of fuel resources for producing energy will depend on such broader trends as the rates of development of industrializing countries and the decisions we made with respect to R & D into energy technology.

The number of important variables involved in the energy picture is so large, and the interconnections among variables so intricate, that common sense and everyday reasoning no longer provide adequate guides to energy policies—if, indeed, they ever did. Nor is there a simple organizational solution of a traditional kind: establishing a federal agency with comprehensive jurisdiction over energy problems, or, alternatively, tinkering with the market mechanism. Agency reorganization is no solution for at least two reasons. First, energy problems cannot be separated neatly from other problems. What would be the relation of a comprehensive jurisdiction over environmental problems? The fragmentation of responsibility for energy policy in the federal government today is a consequence of the intertwining of those problems with others. Second, even if there were such an agency, it too would need a systematic framework within which to take up its decision problems. Tinkering with market mechanisms raises the same difficulty—without a decision framework, we do not know how to tinker.

Hence, the most important organizational requirement for handling energy policy in an intelligent way is the creation of one or more models—either of an optimizing or simulation type—to provide coherence to the decision-making process. No doubt, it is of some importance to locate the responsibility for developing and exploiting such models in appropriate places in the governmental and industrial structure. But the mere existence of the models, wherever located, cannot but have a major impact on energy policy decisions. Surprisingly, the first comprehensive models of the energy system are just now under construction,

although the need for them has been fairly obvious for some years. The tardiness of response to the need is evidence both of the novelty of the modelling technology and the novelty of looking at organization as a collection of decision systems rather than a collection of agencies and departments.⁴

Computer Access to External Information

A third point must be made about the characteristics of the computer as a component of the organization's information processing system. I have mentioned as one limitation of the management information systems of recent years their great reliance on information that is generated internally, within the organization itself—for example, production and accounting information. A major reason for the emphasis on internal information is that, since the organization controls the production of this information, it is not hard to produce it in machine-readable form. Then no costly step is involved in getting it inside the computer.

If we examine the kinds of external information that executives use, we find that a large proportion of it is simply natural language text—the pages of newspapers, trade magazines, technical journals, and so on. Natural language text can, of course, be stored in computer memory after it is translated into some machine-readable form—punched cards, magnetic tape, or the like. Once stored in memory, computer programs can be written to index it automatically and to retrieve information from it in response to inquiries of a variety of kinds.

The only barrier, therefore, to making available to the mechanized components of organizational information systems the same kind of external information that executives now rely upon is the cost of putting the information into machine-readable form. Technologically, the obstacle is not insuperable; it is possible to produce devices that will translate printed text into magnetic tape. The costs of doing this, however, are rather high, and the prospects do not appear bright for reducing them rapidly.

This particular Gordian knot should be cut, not united. Substantially every word that is now printed in a newspaper, journal, or book passes at some time during its prior history through a machine—typewriter or teletyping machine—that could produce a machine-readable version of the text at the same time that it produces the

man-readable version, and at an insignificant incremental cost. Hence, we may look forward to a time in the near future when the written word will be almost universally available in both machine-readable and man-readable editions. Once the switchover begins, so that there is a market for the machine-readable versions, we can expect the conversion process to go very rapidly. It is a little like the telephone—the more people who have them, the more worthwhile to get one.

This development will open up a whole new range of application of computers to organization information systems. It will enable computers to serve as initial filters for most of the information that enters the organization from outside, and will thereby help reduce the attentional demands on executives.

Matching Techniques to Requirements

These comments will serve to indicate what is involved in fitting together the requirements of organization information systems with the characteristics of the information technology that is now available or emerging. The key to the successful design of information systems lies in matching the technology to the limits of the attentional resources. From this general principle, we can derive several rules of thumb to guide us when we are considering adding a component to an existing information-processing system.

In general, an additional component (man or machine) for an information-processing system will improve the system's performance only if:

1. Its output is small in comparison with its input, so that it conserves attention instead of making additional demands on attention;
2. It incorporates effective indexes of both passive and active kinds (active indexes are processes that automatically select and filter information for subsequent transmission);
3. It incorporates analytic and synthetic models that are capable not merely of storing and retrieving information, but of solving problems, evaluating solutions, and making decisions.

These heuristics are applicable to all components of information systems, not just to computers. It is a useful exercise, for example, to look at television in their light, as a component of a political information system. Television can be used to deliver lectures or to exhibit concrete scenes and events. As a source of lectures, its relatively undifferentiated mass audience is a

severe handicap—for almost any individual member of the audience, we could devise a lecture that would be more appropriate than the one actually screened. In communicating information through pictures it has the further disadvantages of its inability to abstract, to generalize, or to sample properly from complex populations of events. It possesses no analytic capabilities other than those provided by the commentator.

To point to these severe deficiencies of television as a source of politically relevant information is not to deny its effects on the political system. The most obvious of these effects is its power to focus the attention of an entire society for a period of time on a particular set of events—whether these be trips to the moon, riots, or summit conferences. In general, television *informs* very little about such events, but it may succeed in evoking strong emotions simultaneously from millions of viewers, and consequently in rearranging the public agenda.

Politics as Information Processing

This last example, as well as the earlier example of energy policy, are illustrative of a broader point: that our political institutions are organizations, and that what we have said about the design of information-processing systems for organizations applies fully to the design of the decision-making components of the political system. Nowhere is the problem of attention management and the conservation of attention of greater importance than in the political process.

The themes of alienation of the electorate, and the need to “return government to the people,” which have been prominent in recent public discussion of our institutions are, of course, not new to the American political scene. Cynicism about the political process is deeply ingrained in the American culture. For the most part, proposals for political reform have focused on strengthening controls over the professional full-time participants in the system, elected or appointed (e.g., recall elections, presidential primary), or on creating new channels of direct popular participation in decision making (e.g., the initiative and referendum, official provision for citizen participation in program administration).

As high technology has come to play a larger role in our society, and in the provision of governmental services in particular, the growing difficulties of assuring that participation will be

informed participation have not gone unnoticed. Informed participation in decision is a problem not only for the part-time contributors to the system—the voters—but for elected representatives and high-level administrators as well. It is at the core of the familiar organizational problem of the relation of experts to laymen.

An Example of Poorly Informed Participation

The phosphate detergent fiasco is only the most recent of innumerable instances of the difficulties of settling technical issues with wide public involvement. A problem existed: excessive algal bloom in lakes. It was known that this condition only incurred in waters having a high level of organic nutrients; phosphate, in particular, was highly suspect as culprit. (Clear scientific evidence on the precise mechanisms of eutrophication, and the precise conditions under which it will or won't occur is nearly nonexistent.) A substantial fraction of the phosphate in water is contributed by household detergents. There exist non-phosphate or low-phosphate detergents. Therefore—but the “therefore,” however obvious it may seem, left unanswered most of the critical questions: Would removal of phosphate from household detergents do any good if other phosphate sources remained? Was phosphate, in fact, the principal culprit? What are the alternative methods, other than changing detergents, for eliminating phosphate from lakes? What are the properties of the alternative detergents, and what consequences would their use have? What would be the relative costs of various courses of action?

As our previous analysis should lead us to predict, the mass media had a large impact in focusing public attention on the eutrophication problem, and in creating insistent demands for forceful action. There is no difficulty in picturing on television a dead fish or (at least on color television) an algae-green lake. What the mass media could not do was to remove the scientific unknowns from the problem, or provide a systems analysis of alternatives. As a result of the pressure for action, the federal government came close to banning phosphate from detergents, but unfortunately had not done so by the time some of the difficulties and dangers of the alternatives had become known.

My intent here is not to suggest what the solution of the problem should have been, but to use these events to point to the weaknesses of our current decision-making processes in political mat-

ters where complex technical issues are involved. Most of the current proposals to remedy these weaknesses are of a traditional kind: create new organizations to provide the technical expertise that is now lacking. The two most prominent proposals are the creation or strengthening of organizations that would represent the consumer and the creation of organizations for technology assessment, these latter to be responsible to the legislative body.

So far as these proposals go, they seem to me sound, but I would warn that we should not expect too much from them. In particular, in talking about "consumer organizations," we must remember that that is precisely what a government is created to be, and thus new organizations of this kind are no more exempt from the iron law of bureaucracy than are existing institutions (as experience with trade unions, political parties, and the American Medical Association should have taught us).

The Contribution of Systems Analysis

If magic is to be performed, it will not come through the mere creation of new organizations. The problem is not primarily one of control but of information—not one of enforcing virtue but of discovering the path of virtue. We do not need new organizations so much as we need new decision processes. To illustrate my meaning, let me cite some happier examples than the phosphate debate of recent public discussion.

The struggles over the anti-ballistic missile and the supersonic transport are, to my mind, examples of what we may hope for in the way of informed discussion of highly technical and complex issues. This does not mean that the correct decisions were necessarily reached. I have no more infallible means for deciding that than did the disputants at the time of the debate. Honest and reasonable men could and did take either side of either question. But what distinguished these particular debates was that both sides were armed with sophisticated analyses based on man-years of careful study supported by quantitative models. For this reason, it was possible for the layman, with a reasonable expenditure of time, to understand where the differences lay—which disagreements about assumptions were responsible for the divergent conclusions reached. Moreover, for each of the decisions there was not a single analysis but several, prepared by protagonists that had different sets of interests and different viewpoints.

Thus, the ABM and SST decision processes were informed by analyses that met three essential criteria: comprehensiveness, technical sophistication, and pluralism. Satisfaction of these criteria does not guarantee that correct decisions are reached—in a world of uncertainty and conflict of interest, that guarantee can seldom be provided. It does tend to assure that the decisions reached are such as reasonable men of good will could arrive at. And it is the guarantee of due process, not the guarantee of infallibility, that democratic institutions require.

In making this plea for more and better systems analyses, and more and better institutional arrangements for carrying them out, I should not want to be thought to be urging centralization of any and all decisions. An equally important direction for improvement of decision processes is to instill new vigor into market mechanisms as an essential component of a decentralized society. Our growing awareness of the externalities, particularly environmental costs, that the price system has ignored in the past has caused something of a crisis of confidence in market mechanisms as social regulators and decision makers. Where externalities are present, there is an alternative, however, to centralization: incorporating the externalities in the pricing process. A tax on oxides of sulphur emitted into the atmosphere is an example of such a measure. To attack such a proposal as "a license to pollute" is to substitute sloganizing for problem solving. We need not only to solve the problems we perceive, but, in solving them, to minimize the drain on our scarce resources. Using the price system as motivator, where it is applicable, is one of the most powerful means available to us for enlisting a multitude of energies and minds into the search for least-cost solutions to our problems.

Conclusion

The major problems of governmental (and corporate and educational) organization today are not problems of departmentalization and coordination of operating units. Instead, they are problems of organizing information storage and information processing—not problems of the division of labor, but problems of the factorization of decision making. These organizational problems are best attacked, at least to a first approximation, by examining the information system in abstraction from agency and department structure.

With the rapid development of information-

processing technology, the corporate and public decision-making processes are becoming immensely more sophisticated and rational than they were in past eras. If we require any proof for this, we need only compare the ABM debate (regardless of whether we like its outcome) with any debate on the Acropolis reported by Thucydides—or, for that matter, with any debate in the U.S. Congress in the first half of this century.

With the development of information-processing technology, we have a growing capacity to consider interactions and tradeoffs among alternatives and consequences; to cumulate our understanding of fragments of the whole problem by embedding these fragments in comprehensive models.

Barbara Ward and others have pointed out to us that the largest crises in our world today are crises of aspirations. The population problem is as old as Man. What is new about it today is that we are resolved not to accept a gloomy outcome, but to deal with it. For centuries, Man's actions have been creating all kinds of unintended and unexpected consequences. He could live in good conscience with his actions to the extent that he was unaware of these consequences. Today, we can trace minute and indirect effects of our behavior: the relation of smoking to cancer, the relation of the brittleness of eagles' eggs to the presence of DDT in the environment. With this new ability to trace effects, we feel responsible for them in a way we previously did not. The intellectual awakening is also a moral awakening.

The new problems created (or made visible) by our new scientific knowledge are symptoms of progress, not omens of doom. They demonstrate that Mankind now possess the analytic tools that are basic to understanding his problems—basic to understanding the human condition.

Of course, to understand problems is not necessarily to solve them. But it is the essential first step. The new information technology that we are creating enables us to take that step.

Some Further Reading

The premises about the nature of organizations upon which this essay rests are developed and documented in *Administrative Behavior* (Macmillan, 1947, 1957) and in the two final chapters of March and Simon, *Organizations* (Wiley, 1958). Fuller discussions of the potentialities of com-

puters and systems design tools in organizations will be found in the last half of *The Shape of Automation* (Harper & Row, 1965) and in chapter 3 of *The Sciences of the Artificial* (M.I.T. Press, 1969).

I have proposed a taxonomy of species of information in my essay "Research for Choice," in W. R. Ewald, Jr. (ed.), *Environment and Policy: The Next Fifty Years* (Indiana University Press, 1968, pp. 360-380), and have drawn out the implications of attention scarcity for technology assessment in "Designing Organizations for an Information-Rich World," published in Martin Greenberger (ed.), *Computers, Communications, and the Public Interest* (Johns Hopkins Press, 1971, pp. 37-52).

Notes

1. It takes a bit of stretching to include military organizations in this category of "production organizations," and regulatory bodies do not really fit at all. However, most of the literature, empirical and theoretical, of organization theory in recent years has focused on industrial organizations and governmental agencies that provide public services, both of which categories do fit. I shall make no attempt here to discuss regulatory agencies. Military organizations, however, have a much larger component of information processing in their total pattern of activity than do, say, typical manufacturing organizations. Hence it is probably no accident that the concern with organizing for effective information processing first made its appearance on a large scale in military organizations, and that they have been and still are the initial testing grounds for most modern information processing technology.
2. The names of Lewin, Roethlisberger, Likert, MacGregor, and Argyris will identify the general range of approaches and emphases within the human relations movement. Many other names could, of course, be added to the list, which is only illustrative.
3. One evidence of the degree of novelty of the computer's capabilities is the resistance it evokes from those who refuse to see in it anything more than an enlarged desk calculator. Not since the Darwinian controversy of the past century have we seen such a passionate defense of the uniqueness of man against claims of kinship by systems that don't belong to his species.
4. We have now had a generation's experience with decision models for economic policy. The construction and testing of such models in the United States has been carried out in considerable part by nongovernmental agencies—the Cowles Foundation for Research in Economics, and The Brookings Institution, for example. Since the day when President Nixon declared himself to be a Keynesian, their impact on government decisions could no longer be in doubt, although the impact certainly preceded that declaration by a decade.

or more. The econometric models have generally used classical analytic mathematical techniques, but the computer has been essential to carrying out the calculations. A somewhat different example is provided by several linear programming models that have been constructed, mainly under university auspices, to guide water policy decisions. In both of these cases we see decision-making systems being designed in relative independence from reorganizations of a traditional kind. It is interesting to speculate whether all of the agency-shuffling reorganizations of federal agencies since 1937 have had as great an effect on public policy as these new decision-making systems.



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